MILITARY SPECIFICATION

MOTOR, ROCKET, METAL PARTS FOR AND LOADING ASSEMBLY FOR DEMOLITION KIT, CRATERING: M180

This specification is approved for use by U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for the metal parts and loading assembly of the rocket motor for Demolition Kit, Cratering: M180.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Box, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
QQ-S-781	-	Strapping, Steel, and Seals.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-117	-	Bag, Sleeve and Tubing, Interior Packaging.
MIL-I-6868	-	Inspection Process, Magnetic Particle.
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-I-48110 - Igniter Assembly, Rocket, for Demolition Kit, Cratering, M180.
MIL-P-50202 - Propellant Charge, Rocket, for Demolition Kit, Cratering, M180.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9329589 - Rocket Motor Assembly.
9329725 - Packing and Marking For Box, Fiberboard For Rocket Motor Assembly or Rocket Motor and Detonator Assembly.
9329726 - Packing and Marking For Box, Packing For Box, Fiberboard For Rocket Motor and Detonator Assembly or Rocket Motor Assembly.
9340544 - Packing and Marking For Box, Packing For Box, Fiberboard For Rocket Motor and Detonator Assembly or Rocket Motor Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 (Part 10) - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM B 117 (Part 10)- Salt Spray (Fog) Testing.

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications and standards.

3.3 Rocket motor assembly. The rocket motor assembly shall comply with all requirements specified on Drawing (Dwg.) 9329589, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.4 Magnetic particle inspection (Motor Assembly Weld). The motor assembly weld shall show no evidence of voids, slag deposits, cracks or crack-like indications when tested as specified in 4.5.3.

3.5 Welding. All welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of good fusion, penetration, and soundness of deposit representative of good welding practice. The process shall be such as to prevent warpage and distortion. There shall be no crack in or near welded areas.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, gouges, dirt, grease, rust, and other foreign matter. After loading, all external surfaces, including cavities, shall be free from explosive. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous, however, a few light scratches not exposing base metal may be permitted. All required marking and stamping shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

Part Description	Drawings	Quantity
Head, Rocket	9244407	5
Plate, Retainer, Rocket	9244408	5
Nozzle, Motor Case	9244439	5
Case, Motor	9244440	5
Rocket Motor Case Assembly	9244441	5
Charge Assembly (Propellant)	9329587	10
Rocket Motor Assembly	9329589	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE: Rocket Motor Assembly		SHEET 1 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY: --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Head, Rocket (Dwg. 9244407)				
	Examination for Defects	5		3.2	4.4.2.1
	Hardness	5(a)		3.2	4.5.1
	Salt Spray Test	5(a/b)		3.2	4.5.2
	Plate, Retainer Rocket (Dwg. 9244408)				
	Examination for Defects	5		3.2	4.4.2.2
	Salt Spray Test	5(a/b)		3.2	4.5.2
	Nozzle, Motor Case (Dwg. 9244439)				
	Examination for Defects	5		3.2	4.4.2.3
	Case, Motor (Dwg. 9244440)				
	Examination for Defects	5		3.2	4.4.2.4

NOTES: (a) This test shall be performed on the samples subjected to the examination for defects.
(b) If any sample shows evidence of corrosion or salt deposits, the sample shall be classed defective.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-M-64016(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Rocket Motor Assembly				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rocket Motor Case Asembly (Dwg. 9244441)				
	Examination for Defects	5		3.2	4.4.2.5
	Hardness	5(a)		3.2	4.5.1
	Salt Spray Test	5(a/b)		3.2	4.5.2
	Magnetic Particle Inspection	5(a/b)		3.4	4.5.3
	Charge Assembly (Propellant) (Dwg. 9329587)				
	Examination for Defects	10		3.2	4.4.2.6
	Rocket Motor Assembly (Dwg. 9329589)				
	Examination for Defects	10		3.2	4.4.2.7

NOTES: (a) This test shall be performed on the samples subjected to the examination for defects.
(b) If any sample shows evidence of corrosion or salt deposits, the sample shall be classed defective.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each lot shall contain:

a. Igniter assemblies of one lot interfix number from one manufacturer. (See 4.4.5)

b. Propellant charge of one lot interfix number from one manufacturer and complying with MIL-P-50202.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Head, Rocket				9244407
					NEXT HIGHER ASSEMBLY
					9329589
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness Test		100%	3.2	4.5.1
102	Salt Spray Test	(a)		3.2	4.5.2
103	Minimum Perfect thread, minimum (min.)		0.40%	3.2	Gage
104	Diameter of O ring groove		0.40%	3.2	Gage
105	True position of O ring groove with datum A		0.40%	3.2	Gage
106	Large outside diameter		0.40%	3.2	Gage
107	Thickness of larger portion		0.40%	3.2	Gage
108	Hole missing		0.40%	3.2	Visual
Minor					
201	Pitch diameter of tapped holes (6)		0.65%	3.2	Gage
202	Minor diameter of tapped holes (6)		0.65%	3.2	Gage
203	True position of tapped holes (6)		0.65%	3.2	Gage
204	Width of O ring groove		0.65%	3.2	Gage
205	Distance between top surface and O-ring groove		0.65%	3.2	Gage
206	Total thickness of head		0.65%	3.2	Gage
207	Protective finish missing or damaged (bare spot)		0.65%	3.2	Visual
208	Poor workmanship		1.0%	3.7	Visual

NOTES: (a) Three (3) rocket heads shall be selected from each lot for this test. Lot formation provisions shall be as specified in the applicable finish specification. If any sample shows evidence of corrosion or salt deposits, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Plate, Retainer, Rocket				9244408
					NEXT HIGHER ASSEMBLY 9329587
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray Test	(a)	-	3.2	4.5.2
102	Outside diameter, maximum (max.)		0.40%	3.2	Gage
103	Thickness of plate		0.40%	3.2	Gage
104	Diameter of propellant charge holes		0.40%	3.2	Gage
105	True position of 15 equally spaced holes		0.40%	3.2	Gage
106	True position of 28 equally spaced holes		0.40%	3.2	Gage
107	True position of 8 equally spaced holes		0.40%	3.2	Gage
108	True position of 21 equally spaced holes		0.40%	3.2	Gage
109	Depth of undercut		0.40%	3.2	Gage
Minor					
201	Chamfer missing on propellant charge holes (72)		0.65%	3.2	Visual
202	Protective finish missing or damaged (bare spot)		0.65%	3.2	Visual
203	Poor workmanship		1.0%	3.7	Visual

NOTES: (a) Three (3) plates shall be selected from each lot for this test. Lot formation provisions shall be as specified in the applicable finish specification. If any sample shows evidence of corrosion or salt deposits, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Nozzle, Motor Case		9244439
			NEXT HIGHER ASSEMBLY 9244441

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Smaller inside diameter		0.40%	3.2	Gage
102	Larger inside diameter		0.40%	3.2	Gage
103	Wall thickness at small end		0.40%	3.2	Gage
104	Wall thickness at larger end		0.40%	3.2	Gage
105	True position of larger inside diameter with datum A		0.40%	3.2	Gage
Minor					
201	Total length		0.65%	3.2	Gage
202	Chamfer at smaller diameter		0.65%	3.2	Gage
203	Poor workmanship		1.0%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Case, Motor		9244440
			NEXT HIGHER ASSEMBLY
			9244441

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of cavity from head end to start of tapered shoulder at nozzle end		0.40%	3.2	Gage
102	Total length		0.40%	3.2	Gage
103	Outside diameter at nozzle end		0.40%	3.2	Gage
104	Inside diameter at nozzle end		0.40%	3.2	Gage
105	Wall thickness		0.40%	3.2	Gage
106	Runout of nozzle end		0.40%	3.2	Gage
107	Inside finish improper (head counterbore)		0.40%	3.2	Visual
Minor					
201	Outside radius		0.65%	3.2	Gage
202	Outside radius from outside diameter (nozzle end to nozzle throat)		0.65%	3.2	Gage
203	Poor workmanship		1.0%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9244441
4.4.2.5	Rocket Motor Case Assembly				NEXT HIGHER ASSEMBLY 9329589
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness test		100%	3.2	4.5.1
102	Magnetic particle inspection		100%	3.2	4.5.3
103	Salt Spray Test	(a)	-	3.2	4.5.2
104	Diameter of retaining ring groove		0.40%	3.2	Gage
105	Width of retaining ring groove		0.40%	3.2	Gage
106	Inside diameter of motor case		0.40%	3.2	Gage
107	True position of head end I.D. with datum A		0.40%	3.2	Gage
108	Inside diameter at head end of Motor Case		0.40%	3.2	Gage
109	Outside diameter of Motor Case		0.40%	3.2	Gage
110	Runout of throat diameter with datum A		0.40%	3.2	Gage
111	Diameter of throat		0.40%	3.2	Gage
112	Outside diameter of flange at head end		0.40%	3.2	Gage
113	Runout of nozzle end I.D. with datum A		0.40%	3.2	Gage
114	Weld workmanship unsatisfactory		0.40%	3.5	Visual

NOTES: (a) Three (3) assemblies shall be selected from each lot for this test. Lot formation provisions shall be as specified in the applicable finish specification. If any assembly shows evidence of corrosion or salt deposits, the lot shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	Rocket Motor Case Assembly				9244441
					NEXT HIGHER ASSEMBLY
					9329589
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth of slot in flange from head end		0.65%	3.2	Gage
202	10° and 15° chamfers missing on inside diameter of head end		0.65%	3.2	Visual
203	Radius at forward edge of retaining ring groove missing		0.65%	3.2	Visual
204	Total length		0.65%	3.2	Visual
205	Welding nut (3) missing (when applicable)		0.65%	3.2	Visual
206	Protective finish missing or improper (bare spot)		0.65%	3.2	Visual
207	Poor workmanship		1.0%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Charge Assembly (Propellant)				9329587
					NEXT HIGHER ASSEMBLY 9329589
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant grain missing or damaged (broken, chipped, crack)		0.40%	3.2	Visual
102	Retainer plate damaged (crack or deformed)		0.40%	3.2	Visual
103	Stud missing or damaged		0.40%	3.2	Visual
104	Part incorrectly assembled		0.40%	3.2	Visual
Minor					
201	Rubber band missing		0.65%	3.2	Visual
202	Poor workmanship		1.0%	3.7	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Rocket Motor Assembly				9329589
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance from Retainer to edges of potting compound		0.40%	3.3	Gage
102	O-ring on head missing		0.40%	3.3	Visual
103	Rubber band not removed from Charge Assembly		0.40%	3.3	Visual
104	Any component missing or improperly assembled (Igniter Assembly, igniter retainer, preformed packing, retaining ring (2) rocket head)		0.40%	3.3	Visual
105	Sealant missing from specified surfaces (igniter end)		0.40%	3.3	Visual

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9329589
4.4.2.7	Rocket Motor Assembly				NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Lubricant around O-ring and adjacent head surface missing or improper (3 places)		0.65%	3.3	Visual
202	Marking missing, incorrect, or illegible		0.65%	3.3	Visual
203	Poor workmanship		1.0%	3.7	Visual

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MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Box, Fiberboard (prior to sealing)	1 OF 1	9329725
			NEXT HIGHER ASSEMBLY: 9329726

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Component improperly packed		0.40%	3.2	Visual
Minor 201	Packing components missing, inadequate or improperly positioned		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Box, Fiberboard (Sealed)		9329725
			NEXT HIGHER ASSEMBLY 9329726

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box punctured, torn or cut		0.40%	3.2	Visual
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Visual
203	Sealing strip incomplete or loose		0.65%	3.2	Visual
204	Corner not blunted		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Packing Box (prior to securing top)	1 OF 1	9329726
			NEXT HIGHER ASSEMBLY: -

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents positioned improperly or missing		0.40%	3.2	Visual/Manual
Minor					
201	Foreign material in box		0.65%	3.2	Visual
202	Insufficient area for subsequent seals		0.65%	3.2	Visual
203	End, side, and top fillers missing or inadequate		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-64016(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.11	Wooden Packing Box (Sealed)	1	1	9329726

NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardware missing, broken or loose		0.40%	3.2	Visual/Manual
102	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
103	Contents exposed		0.40%	3.2	Visual
104	Board broken or split		0.40%	3.2	Visual
105	Protruding nail		0.40%	3.2	Visual
Minor					
201	Hardware improperly engaged		0.65%	3.2	Visual/Manual
202	Handle missing, insecure or damaged (deterioration, cut, etc.)		0.65%	3.2	Visual/Manual
203	Contents loose (when box is tilted)		0.65%	3.2	Visual/Manual
204	Strapping improperly assembled		0.65%	3.2	Visual
205	Marking misleading or unidentifiable (including incorrect color code)		0.65%	3.2	Visual
206	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraph (see 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.4.5 Check test for deterioration of igniter assemblies (See MIL-I-48110). If the total time between original acceptance of any igniter assembly lot and the assembly of that lot into the rocket motor exceeds two years, or if the igniters have been subjected to adverse conditions, however brief, at any time since previous tests, the igniter lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-I-48110 immediately before the igniter lot is assembled into the rocket motor. This test shall be performed by the contractor on igniters selected by the Government Inspector at the facility assembling the igniters into the rocket motors. (see 6.3)

4.5 Methods of inspection.

WARNING. This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Hardness. This test shall be performed as specified in ASTM E 18. If any rocket head or rocket motor case assembly fails to comply with the specified hardness requirement, it shall be classed defective and removed from the lot.

4.5.2 Salt spray test. The salt spray test shall be performed utilizing the equipment, methods and procedures specified in ASTM B-117. Exposure time and percent of sodium chloride solution shall be in accordance with the applicable finish specification.

4.5.3 Magnetic particle inspection. The procedures and equipment for performing this inspection shall be in accordance with MIL-I-6868 and utilizing the continuous, wet, fluorescent method. If any rocket motor case assembly fails to meet the requirement specified, it shall be classed defective and removed from the lot. Personnel performing this inspection shall be qualified in accordance with MIL-STD-410.

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5. PACKAGING

5.1 Cleaning. The following items shall be cleaned in accordance with process C-1 of MIL-P-116:

Head, Rocket	Dwg. 9244407
Plate, Retainer, Rocket	Dwg. 9244408
Nozzle, Motor Case	Dwg. 9244439
Case, Motor	Dwg. 9244440
Rocket Motor Case Assembly	Dwg. 9244441
Rocket Motor Assembly	Dwg. 9329589

5.2 Drying. Components referenced in 5.1. shall be dried in accordance with MIL-P-116.

5.3 Preservation.

5.3.1 The Nozzle, Motor Case shall be preserved with P-6 or P-9 preservative in accordance with MIL-P-116.

5.3.2 The Case, Motor shall be preserved with P-2 or P-6 preservative in accordance with MIL-P-116.

5.4 Unit packaging.

5.4.1 Level A. All items referenced in 5.1, except the Rocket Motor Assembly, shall be packaged sub-method IA-8 of MIL-P-116 utilizing a heat sealed barrier bag conforming to Type 1, Class E, of MIL-B-117.

5.4.2 Level B. Package as specified in Level A.

5.4.3 Level C. The Rocket Motor Assembly shall be unit-packaged in accordance with Dwg. 9329725.

5.4.4 Commercial. Commercial packaging shall be in accordance with MIL-STD-1188, except the Rocket Motor Assembly shall not be packaged commerical.

5.5 Intermediate packaging.

5.5.1 Level A. All items referenced in 5.1, except the Rocket Motor Assembly, shall be packaged in a snug-fitting fiberboard box conforming to Class WWVR, Grade V3c, per PP-B-636, utilizing enough cushioning material, PPP-C-843, to assure a tight pack, and closed in accordance with PPP-B-636.

5.5.2 Level B. Intermediate packaging shall be as specified in 5.5.1.

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5.5.3 Commercial. Commercial packaging shall be in accordance with MIL-STD-1188. The Rocket Motor Assembly shall not be packaged commercial.

5.6 Packing.

5.6.1 Level A. Intermediate packages specified in 5.5.1 shall be overpacked in Grade B, Style A, B, I or J, PPP-B-601 boxes; Class 2, Style 4, PPP-B-621 boxes; or Class 3, Style 2 or 3, PPP-B-585 boxes. Closure and strapping shall be in accordance with applicable container specifications except strapping shall conform to Type I or IV, Finish A, QQ-S-781.

5.6.2 Level B. Level B packing shall be as specified in 5.6.1.

5.6.3 Level C. The Rocket Motor Assembly shall be packed in accordance with Dwg. 9329726 or 9340544, as applicable.

5.6.4 Commercial. Commercial packaging shall be in accordance with MIL-STD-1188, except the Rocket Motor Assembly shall not be packed commercial.

5.7 Marking.

5.7.1 Level A. Marking shall be in accordance with MIL-STD-129.

5.7.2 Level B. Marking shall be in accordance with MIL-STD-129.

5.7.3 Level C. Marking shall be in accordance with MIL-STD-129, Dwg. 9329726 or 9340544, as applicable.

5.7.4 Commercial. Marking shall be in accordance with MIL-STD-1188.

5.8 Shipping. When assemblies from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Ordering data. See MIL-A-48078.

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and tests paragraph number.

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6.3 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the igniter assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the igniter assemblies or to the contractor assembling the igniter assemblies into the rocket motors; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the igniter assemblies lot was under the jurisdiction of either contractor.

6.4 Data cards. (See MIL-A-48078). Distribution of data cards shall include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

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